NORTHLIGHT RESEARCH

# The State of the Page

*How a 30-year-old format keeps getting better*

ANNUAL REPORT · 2026 EDITION

Not for redistribution — a sample document

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# Executive summary

KEY MESSAGE

Document technology rarely makes headlines, yet it quietly carries contracts, invoices, manuals and archives for billions of people. This report reviews how a mature page-description format stays reliable across decades of devices, operating systems and software generations.

Three forces explain its longevity: a small, fully specified imaging model; a conservative file structure that tolerates partial damage; and an ecosystem of independent implementations that keep each other honest. Each is examined in the sections that follow.

The findings are drawn from a corpus of 4,812 public documents and from the behaviour of four independent renderers. Where results differ we say so, and where a difference is a bug we link it to the clause of the specification that decides the matter.

> *“Longevity is a design choice, not an accident of adoption.”*
>
> — Dr. Imani Okafor, lead author

| 4,812 | 97.3 % | 12 | 30 yrs |
|---|---|---|---|
| documents analysed | rendered identically | format clauses clarified | of backward compatibility |

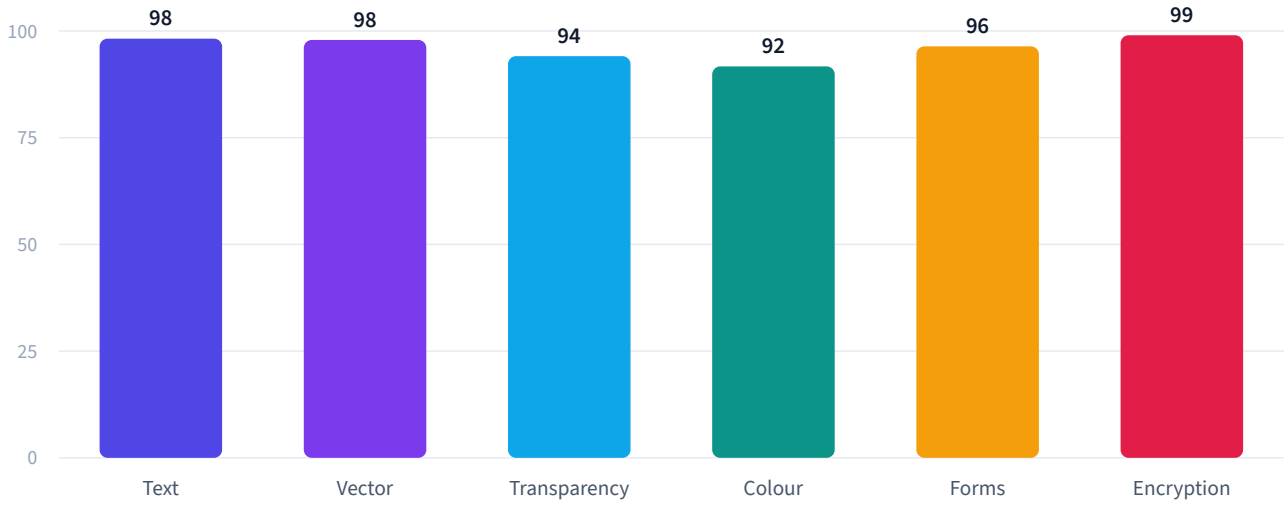
# Findings

Compatibility improved across every feature family we measured. The table lists the share of corpus documents that used each feature and the share rendered without visible difference.

| FEATURE FAMILY | USAGE | IDENTICAL |
|---|---|---|
| Text and fonts | 99.1 % | 98.2 % |
| Vector graphics | 87.4 % | 97.9 % |
| Transparency | 41.6 % | 94.1 % |
| Colour management | 33.8 % | 91.7 % |
| Forms and annotations | 22.5 % | 96.4 % |
| Encryption | 9.3 % | 99.0 % |

FIGURE 1

**Identical rendering by feature family, %**

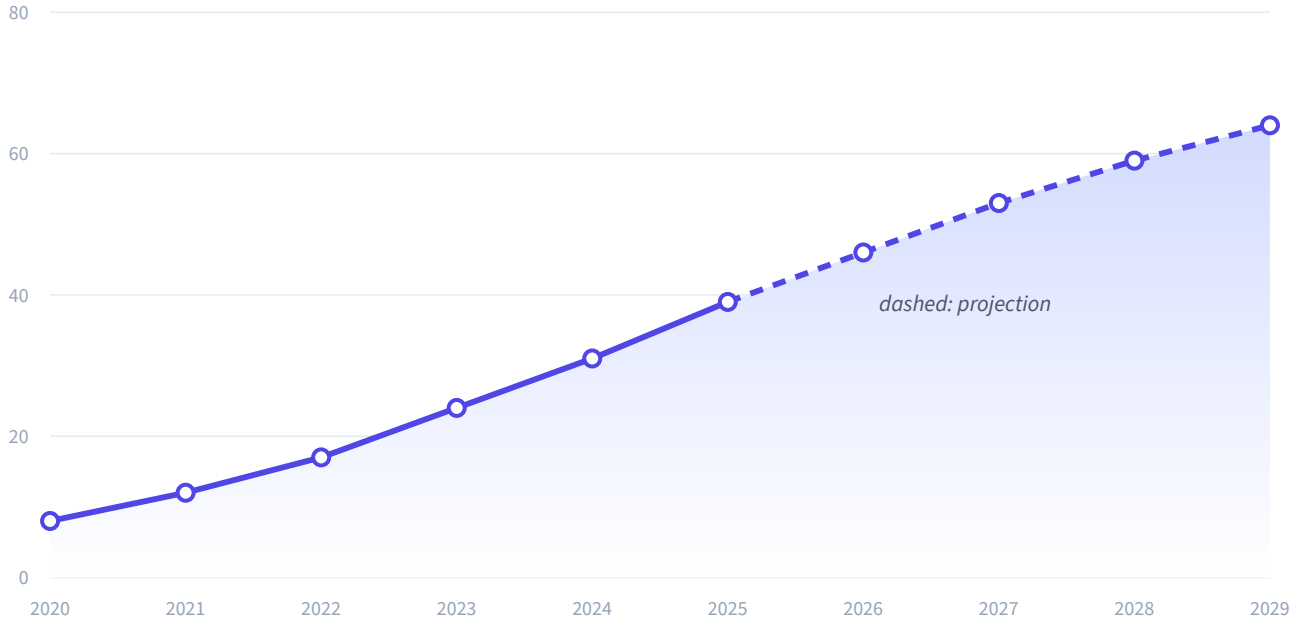


# Outlook

Adoption of tagged, accessible documents is the clearest growth story. If the current trend holds, a majority of new documents will carry a structure tree by 2029.

FIGURE 2

**Share of new documents with a structure tree, %**




### Accessible by default

Authoring tools tag structure automatically.

### Smaller files

Object streams and subsetting cut size by a third.

### Safer by design

AES-256 is now the baseline for protected files.

# Method and sources

Documents were sampled from public repositories, converted to a common baseline where necessary, and rendered at 96 dpi by four independent engines. A document counts as identical when no channel differs by more than 2 % over more than 2 % of pixels.

## REFERENCES


| | |
|---|---|
| ISO 32000-2:2020 | Document management — Portable document format — Part 2: PDF 2.0 |
| ISO 14289-1:2014 | PDF/UA-1 — accessible PDF |
| ISO 19005-4:2020 | PDF/A-4 — archival |
| RFC 8118 | The application/pdf media type |


## Colophon

Set in Source Serif 4 and Source Sans 3 (embedded CFF subsets). Charts are drawn with path operators and shadings; there are no raster images in this file.